illuminati - 2020

Advanced Physics Assignment-1

Electrostatics, DC Circuits, Capacitors, Magnetism, EMI, AC Circuits

SECTION-1	SINGLE CORRECT ANSWER TYPE

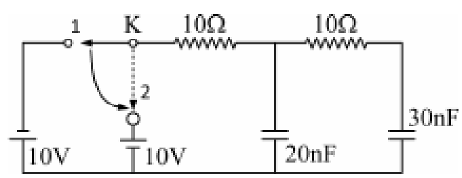
1. 'n' identical charge particle are placed on the vertices of a regular polygon of 'n' sides of side length 'a'. One of the charge particles is released from polygon. When this particle reaches a far off distance, another particle adjacent to the first particle is released. The difference of kinetic energies of both the particles at infinity is k. Magnitude of charge is :

(A) $\sqrt{4\pi\varepsilon_0 ak}$ **(B)** $\dfrac{k}{4\pi\varepsilon_0 a}$ **(C)** $\dfrac{k}{a}$ **(D)** $\sqrt{ka}$

2. A cell of emf $E = 1.5\ V$ and internal resistance $r = 0.5\Omega$ is connected to a non ohmic resistor in which the current varies with the voltage as $I = V^2$. Calculate the current drawn from the cell and its terminal voltage.

(A) 1 volt, 1A **(B)** 2 volt, 1A **(C)** 2 volt, 2A **(D)** 1 volt, 2A

3. In the given circuit, when the switch is shifted from position 1 to position 2, find the total heat dissipated in the resistors.



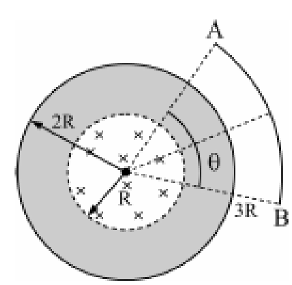
(A) $10\mu J$ **(B)** $20\mu J$

(C) $5\mu J$ **(D)** Zero

4. A particle of mass $m = 3\times10^{-19} kg$ and charge $q = 2\times10^{-12} C$ is moving with a uniform velocity along x-axis in a region of uniform electric field $E_z = 5KV/m$ and magnetic field $B = 40\ mT$ exists mutually perpendicular to each other. The angle between the $\vec{B}$ field and the +x-axis is $\theta = 30°$. The pitch of the trajectory of the particle after removing the electric field is nearly:

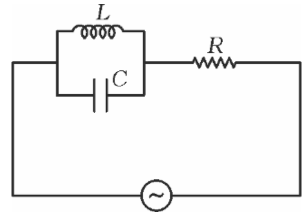
(A) 0.17 cm **(B)** 3.93 m **(C)** 5 m **(D)** 17.73 m

5. In the given figure, there are two concentric cylindrical region in which time varying magnetic field is present. From the centre to radius R magnetic field is perpendicular in to the plane varying as $\dfrac{dB}{dt} = 2k$ and in a region from R to $2R$ magnetic field is perpendicular out of the plane varying as $\dfrac{dB}{dt} = 4k$. Find the induced emf across an arc AB of radius $3R$.



(A) $6R^2k\theta$ **(B)** $5R^2k\theta$ **(C)** $7R^2k\theta$ **(D)** None of these

6. In the given circuit, $L = \frac{50}{\pi^2} \times 10^{-2}$ H, $C = 200\mu F$ and $R = 100\Omega$. An alternating voltage $V = 5 \sin 100\pi t$ is applied across the circuit. Find current in the resistance R and voltage across inductor as a function of time t.



(A) $0A,\ v = 5\sin 100\pi t$ **(B)** $I = 0.05\sin 100\pi t,\ v = 5\sin 100\pi t$

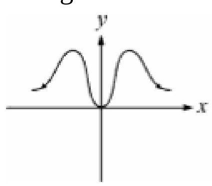
(C) $I = 0.05\sin 100\pi t,\ v = 0$ volt **(D)** None of these

7. An ideal battery of electromotive force ξ is connected in series with an ammeter and a voltmeter of unknown resistances. If a certain resistance is connected in parallel with the voltmeter, the voltmeter and the ammeter readings becomes $1/\eta$ and η times of their respective initial readings. What is the initial reading of the voltmeter?

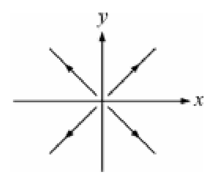
(A) $\frac{\xi}{(\eta+1)}$ **(B)** $\frac{\eta\xi}{(\eta+1)}$ **(C)** $\frac{\eta\xi}{(\eta-1)}$ **(D)** $\frac{(\eta+1)\xi}{\eta}$

8. In a certain region of space, the potential field depends on x and y coordinates as $V = (x^2 - y^2)$. The corresponding electric field lines in x-y plane are correctly represented by :

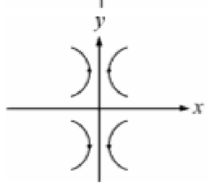
(A)



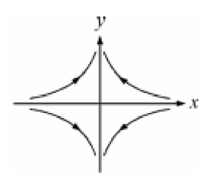
(B)



(C)



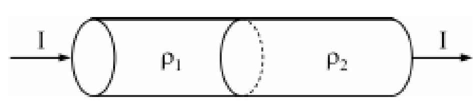
(D)



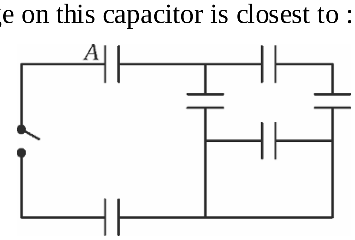
9. Two long straight cylindrical conductors with resistivity ρ_1 with ρ_2 respectively are joined together as shown in figure. The radius of each of the conductor is 'a'. If a uniform total current I flows through the conductors, the total free charge at the interface of the two conductors is:

(A) zero **(B)** $\frac{(\rho_1 - \rho_2) I \varepsilon_0}{2}$

(C) $\varepsilon_0 I(\rho_2 - \rho_1)$ **(D)** $\varepsilon_0 I(\rho_1 - \rho_2)$

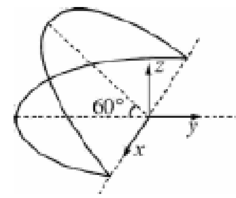


10. In the circuit shown, all capacitors are identical, initially the switch is open and only the capacitor A is charged. After the switch is closed and a steady-state is established, charge on the capacitor A becomes $5.0\mu C$. Initial charge on this capacitor is closest to :



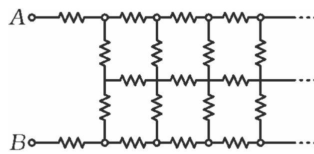
(A) 7.5 μC **(B)** 8.0 μC **(C)** 13.3 μC **(D)** 15.0 μC

11. A circular loop of wire of radius a is bent about its diameter such that the angle between the plane of two halves become 60°. The resistance per unit length of the wire is r_0. The magnetic field is varying as $B = B0t$ along the (+) Z-axis. Find the total charge flown through the loop in first t seconds.



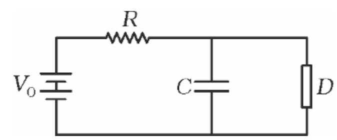
(A) $\frac{B_0at}{2r_0}$ **(B)** $\frac{B_0at}{8r_0}$ **(C)** $\frac{B_0at}{r_0}$ **(D)** $\frac{2B_0at}{r_0}$

12. The circuit shown in the diagram extends to the right upto infinity. Each branch resistance is denoted by r. What is the resistance between the terminals A and B?



(A) $\frac{(\sqrt{5}+1)r}{2}$ **(B)** $\frac{(\sqrt{5}-1)r}{2}$ **(C)** $(\sqrt{5}+1)r$ **(D)** $(\sqrt{5}-1)r$

13. In the circuit shown, the device D has a property that, if initially non-conducting, it remains non-conducting until voltage across it rises to a value $V_1(<V_0)$. It then rapidly discharges the capacitor until the voltage across it drops to a negligibly small value, where upon it returns to a non-conducting state. The voltage developed across the capacitor is :



(A) periodic with time period RC

(B) periodic with time period $RC\ln\left(\frac{V_0}{V_1}\right)$

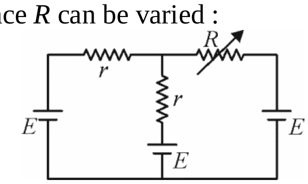
(C) periodic with time period $RC\ln\left(\frac{V_0}{V_0-V_1}\right)$

(D) periodic with time period $RC\ln\left(\frac{V_0^2}{V_1(V_0-V_1)}\right)$

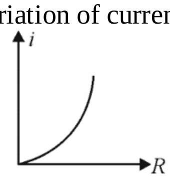
14. Two identical conducting spheres each of radius a are placed at centre to centre separation $d(d >> a)$. They are kept in a homogeneous medium of permittivity ε and resistivity ρ. Which one of the following is a correct expression of resistance between them?

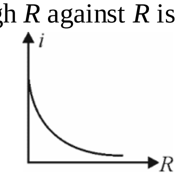
(A) $\frac{\rho}{2\pi a}$ **(B)** $\frac{\rho}{4\pi a}$ **(C)** $\frac{\rho d}{2\pi a^2}$ **(D)** $\frac{\rho d}{4\pi a^2}$

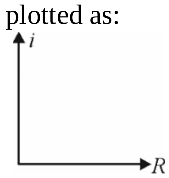
15. In the shown circuit the resistance R can be varied :

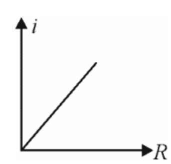


The variation of current through R against R is correctly plotted as:

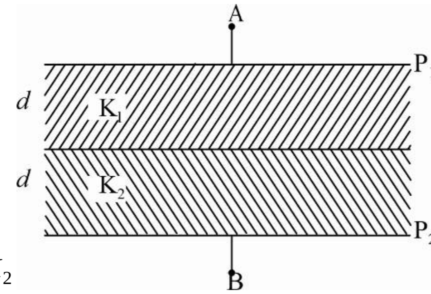
(A) 

(B) 

(C) 

(D) 

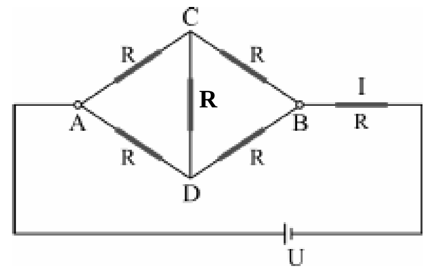
16. In the figure shown P_1 and P_2 are two conducting plates having charges of equal magnitude and opposite sigh. Two dielectrics of dielectric constant K_1 and K_2 fill the space between the plates as shown in the figure. The ratio of electrical energy in 1st dielectric to that in the 2nd dielectric is:



(A) 1:1 **(B)** $K_1 : K_2$

(C) $K_2 : K_1$ **(D)** $K_2^2 : K_1^2$

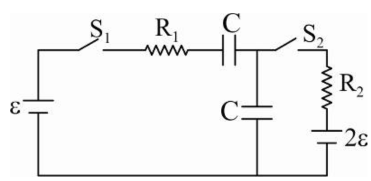
17. If the values of the resistance of the resistors shown in the figure are equal then the current in the main branch is I. By what factor does this current change if the resistance of the two resistors, which are diagonally opposite each other, is doubled ?



(A) 1 : 1 **(B)** 2 : 3

(C) 6 : 5 **(D)** 5 : 6

18. In the circuit shown, switch S_2 is closed first and is kept closed for a long time. Now S_1 is closed. Just after that instant the current through S_1 is:



(A) $\frac{\varepsilon}{R_1}$ towards right

(B) $\frac{\varepsilon}{R_1}$ towards left

(C) Zero

(D) $\frac{2\varepsilon}{R_1}$ towards right

19. When 100V DC is applied across a solenoid, a current of 1 A flows in it. When 100V AC is applied across the same coil, the current drops to 0.5A. If the frequency of the AC source is 50 Hz, the impedance and inductance of solenoid are :

(A) 100Ω, 0.93H **(B)** 200Ω, 1.0H **(C)** 10Ω, 0.86H **(D)** 200Ω, 0.55H

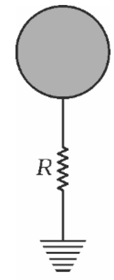
20. Two ammeters, 1 and 2, have different internal resistances : r1 (known) and r2 (unknown). Each ammeter has an analog scale such that the angular deviation of the needle from zero is proportional to the current. Initially, the ammeters are connected in series and then to a source of constant voltage. The deviations of the needles of the ammeters are θ1 and θ2, respectively. The ammeters are then connected in parallel and then to the same voltage source. This time, the deviations of the needles are θ_1' and θ_2' respectively. Find $r2$.

(A) $r_2 = r_1 \frac{\theta_1\theta_2}{\theta_1'\theta_2'}$ **(B)** $r_2 = r_1 \frac{\theta_1\theta_1'}{\theta_2\theta_2'}$ **(C)** $r_2 = r_1 \frac{\theta_2\theta_1'}{\theta_1\theta_2'}$ **(D)** $r_2 = r_1 \frac{\theta_1\theta_2'}{\theta_2\theta_1'}$

SECTION-2	LINK-COMPREHENSION TYPE

PARAGRAPH FOR QUESTIONS 21 - 22

A conducting balloon of radius a is charged to a potential V_0 and held at a large height above the earth surface. The large height of the balloon from the earth ensures that charge distribution on the surface of the balloon remains unaffected by the presence of the earth. It is earthed through a resistance R and a valve is opened. The gas inside the balloon escapes through the valve and the size of the balloon decreases. The rate of decrease in radius of the balloon is controlled in such a manner that potential of the balloon remains constant. Assume electric permittivity of the surrounding air equal to that of free space (ε_0) and charge does not leak to the surrounding air.



21. Time rate at which radius r of the balloon changes is best represented by the expression :

(A) $\frac{1}{4\pi\varepsilon_0 R}$ **(B)** $-\frac{1}{4\pi\varepsilon_0 R}$ **(C)** $\frac{r}{4\pi\varepsilon_0 aR}$ **(D)** $-\frac{r}{4\pi\varepsilon_0 aR}$

22. How much heat is dissipated in the resistance R until radius of the balloon becomes half?

(A) $0.5\pi\varepsilon_0 aV_0^2$ **(B)** $\pi\varepsilon_0 aV_0^2$ **(C)** $2\pi\varepsilon_0 aV_0^2$ **(D)** $4\pi\varepsilon_0 aV_0^2$

PARAGRAPH FOR QUESTIONS 23 - 25

In a series L-R circuit, connected with a sinusoidal ac source, the maximum potential difference across L and R are respectively 3 volts and 4 volts.

23. At an instant the potential difference across resistor is 2 volts. The potential difference in volt, across the inductor at the same instant will be :

(A) $3\cos 30°$ **(B)** $3\cos 60°$ **(C)** $3\cos 45°$ **(D)** None of these

24. At the same instant, the magnitude of the potential difference in volt, across the ac source may be :

(A) $4+3\sqrt{3}$ **(B)** $\frac{4+3\sqrt{3}}{2}$ **(C)** $1+\frac{\sqrt{3}}{2}$ **(D)** $2+\frac{\sqrt{3}}{2}$

25. If the current at this instant is decreasing the magnitude of potential difference at that instant across the ac source is:

(A) Increasing **(B)** Decreasing **(C)** Constant **(D)** Cannot be said

PARAGRAPH FOR QUESTIONS 26 - 28

It is possible to take a high quality photograph of a very fast moving object by illuminating the object for quite a small fraction of a second. You may have come across photographs of a bullet penetrating a banana or an apple in many text books or magazines. This is called ‘Stop Action’ photography because the ‘fast moving object travels a very short distance during the time of illumination. Harold Edgerton, the inventor of stroboscope, was a pioneer of this kind of photography. A normal photographic plate works properly if it receives an energy of $4J$ during the exposure. To release this energy in a very small fraction of time, huge amount of power is required. Such huge power can not be generated directly from a battery because of its high internal resistance. To produce such power a capacitor is used. The time in which a capacitor discharges can be very short. Although, theoretically it would take a long time for a capacitor to discharge completely, it discharges almost completely in about 10 time constants. Consider the following situation. A capacitor of $200\,\mu F$, storing $4J$ energy is made to discharge through a flash light in 2 ms. The Time constant of this circuit is 0.2 ms. This setup is used to take a picture of a bullet moving at a speed of 100 m/s. Assume that the flash light acts as a resistor and there is no other resistance in the circuit.

26. The image of the moving bullet is formed on the photographic plate through a converging lens. If we use a lens of power 10 diopters, the lens to photographic plate distance is 15 cm and the bullet moves perpendicular to the principal axis, what is the distance covered by bullet as seen on photographic plate ?

(A) 1 cm **(B)** 5 cm **(C)** 10 cm **(D)** 20 cm

27. What is the order of energy delivered to the flash light in 0.2 ms (approx.) ?

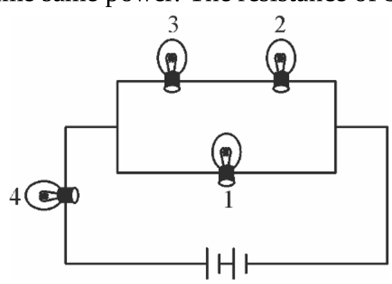
(A) 0.4 J **(B)** 1.83 J **(C)** 2.74 J **(D)** 3.45 J

28. What is the initial current in the circuit

(A) 200 A **(B)** 120 A **(C)** 700 A **(D)** 3700 A

PARAGRAPH FOR QUESTIONS 29 - 31

All bulbs consume same power. The resistance of bulb 1 is 36Ω.



29. What is the resistance of bulb 3 ?

(A) 4Ω **(B)** 9Ω **(C)** 12Ω **(D)** 18Ω

30. What is the resistance of bulb 4 ?

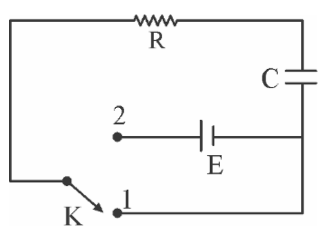
(A) 4Ω **(B)** 9Ω **(C)** 12Ω **(D)** 18Ω

31. What is the voltage output of the battery if the power of each bulb is 4 W ?

(A) 12 V **(B)** 16 V **(C)** 24 V **(D)** None of these

PARAGRAPH FOR QUESTIONS 32 - 34

In the shown circuit involving a resistor of resistance $R\ \Omega$, capacitor of capacitance C farad and an ideal cell of emf E volts, the capacitor is initially uncharged and the key is in position 1. At $t = 0$ second, the key is pushed to position 2 for $t_0 = RC$ seconds and then key is pushed back to position 1 for $t_0 = RC$ seconds. This process is repeated again and again. Assume the time taken to push key from position 1 to 2 and vice versa to be negligible.



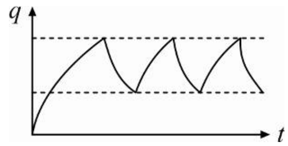
32. The charge on capacitor at $t = 2RC$ second is :

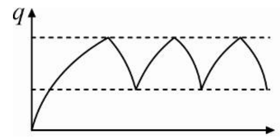
(A) CE **(B)** $CE\left(1-\frac{1}{e}\right)$ **(C)** $CE\left(\frac{1}{e}-\frac{1}{e^2}\right)$ **(D)** $CE\left(1-\frac{1}{e}+\frac{1}{e^2}\right)$

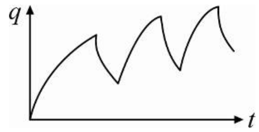
33. The current through the resistance at $t = 1.5\,RC$ seconds is :

(A) $\frac{E}{e^2R}\left(1-\frac{1}{e}\right)$ **(B)** $\frac{E}{eR}\left(1-\frac{1}{e}\right)$ **(C)** $\frac{E}{R}\left(1-\frac{1}{e}\right)$ **(D)** $\frac{E}{\sqrt{e}R}\left(1-\frac{1}{e}\right)$

34. Then the variation of charge on capacitor with time is best represented by :

(A) 

(B) 

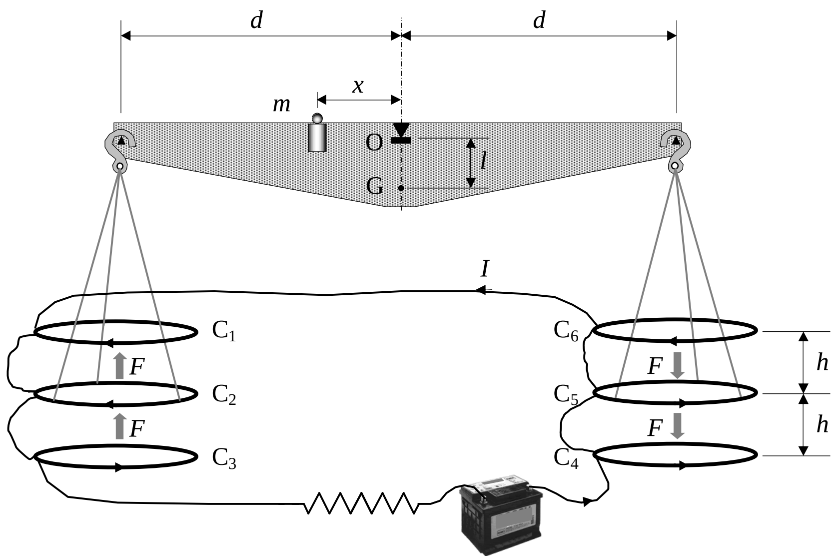
(C) 

(D) 

PARAGRAPH FOR QUESTIONS 35 - 37

Determination of the ampere

Passing a current through two conductors and measuring the force between them provides an absolute determination of the current itself. The "Current Balance" designed by Lord Kelvin in 1882 exploits this method. It consists of six identical single turn coils $C_1 ... C_6$ of radius *a*, connected in series. As shown in figure, the fixed coils $C_1, C_3, C_4,$ and C_6 are on two horizontal planes separated by a small distance 2*h*. The coils C_2 and C_5 are carried on balance arms of length *d*, and they are, in equilibrium, equidistant from both planes. The current *I* flows through the various coils in such a direction that the magnetic force on C_2 is upwards while that on C_5 is downwards. A mass *m* at a distance *x* from the fulcrum *O* is required to restore the balance to the equilibrium position described above when the current flows through the circuit. Let *M* be the mass of the balance (except for *m* and the hanging parts), *G* its centre of mass and *l* the distance $\overline{OG}$.



35. Compute the force *F* on C_2 due to the magnetic interaction with C_1. For simplicity assume that the force per unit length is the one corresponding to two long, straight wires carrying parallel currents.

(A) $\dfrac{\mu_0 I^2}{2\pi h}$ (B) $\dfrac{\mu_0 I^2 a}{h}$ (C) $\dfrac{2\mu_0 I^2 a}{h}$ (D) $\dfrac{\mu_0 I^2}{ah}$

36. The current *I* is measured when the balance is in equilibrium. Give the value of *I* in terms of the physical parameters of the system. The dimensions of the apparatus are such that we can neglect the mutual effects of the coils on the left and on the right.

(A) $\left(\dfrac{mghx}{4\mu_0 ad}\right)^{1/2}$ (B) $\left(\dfrac{mghx}{8\mu_0 ad}\right)^{1/2}$ (C) $\left(\dfrac{mghx}{2\mu_0 ad}\right)^{1/2}$ (D) $\left(\dfrac{mgax}{4\mu_0 hd}\right)^{1/2}$

37. The balance equilibrium is stable against deviations producing small changes δ_z in the height of C_2 and $-\delta_z$ in C_5. Compute the maximum value $\delta_{z_{\max}}$ so that the balance still returns towards the equilibrium position when it is released. (Consider that the coils centres remain approximately aligned)

(A) $\dfrac{M\ell h^2}{2mxd}$ **(B)** $\dfrac{M\ell h^2}{mxd}$ **(C)** $\dfrac{M\ell h}{mxd}$ **(D)** $\dfrac{M\ell h}{mx}$

PARAGRAPH FOR QUESTIONS 38 - 40

Method of images

A point charge q is placed in the vicinity of a grounded metallic sphere of radius R [see figure (i)], and consequently a surface charge distribution is induced on the sphere. To calculate the electric field and potential from the distribution of the surface charge is a formidable task. However, the calculation can be considerably simplified by using the so called method of images. In this method, the electric field and potential produced by the charge distributed on the sphere can be represented as an electric field and potential of a single point charge q' placed inside the sphere (you do not have to prove it). Note : The electric field of this image charge q' reproduces the electric field and the potential only outside the sphere (including its surface).

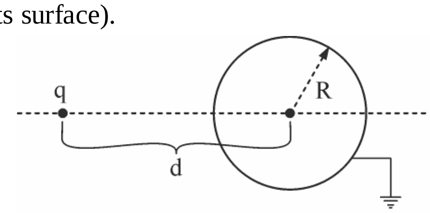


Figure (i)

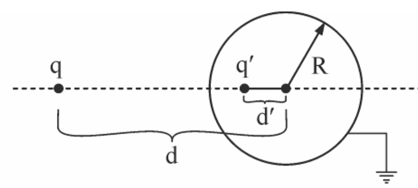


Figure (ii)

The symmetry of the problem dictates that the charge q' should be placed on the line connecting the point charge q and the centre of the sphere [see figure (ii)]. The value of q' and its distance from the centre of sphere d' can be found by using the fact that potential at all points of the sphere must be zero.

38. Value of q' is :

(A) $q\dfrac{R}{d}$ **(B)** $q\dfrac{d}{R}$ **(C)** $-q\dfrac{R}{d}$ **(D)** $-q$

39. Value of d' is :

(A) d **(B)** $\dfrac{R^2}{d}$ **(C)** $\dfrac{d^2}{R}$ **(D)** $\dfrac{2R^2}{d}$

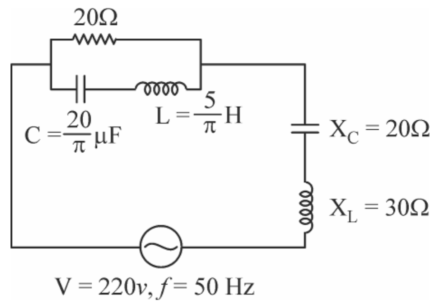
40. Find the magnitude of force acting on charge q.

(A) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q^2Rd}{(d^2-R^2)^2}$ **(B)** $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q^2}{(d^2-R^2)}$

(C) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{q^2d^2}{(d^2-R^2)^2}$ **(D)** zero

SECTION-3 | **MULTIPLE CORRECT ANSWER TYPE**

41. The circuit shown below is :



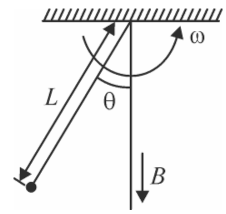
(A) purely resistive

(B) purely inductive

(C) purely capacitive

(D) zero power factor

42. A charged particle having its charge to mass ratio as β goes in a conical pendulum of length L making an angle θ with vertical and angular velocity ω. If a magnetic field B is directed vertically downwards (see figure), then choose correct options.



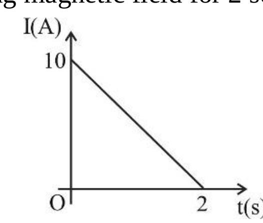
(A) $B = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L \cos\theta}\right]$

(B) Angular momentum of the particle about the point of suspension remains constant

(C) If the direction of B were reversed maintaining same ω and L, then θ will remain unchanged

(D) Rate of change of angular momentum of the particle about the point of suspension is not a constant vector

43. A variable current flows through a 10 Ω resistor coil kept in changing magnetic field for 2 seconds.



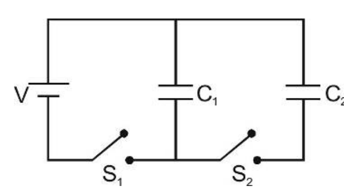
(A) Change in magnetic flux is 100Wb.

(B) Rate of change of magnetic flux is decreasing

(C) Total heat produced in the resistor is 666.67 J.

(D) Maximum power during the flow of current is 1000 W.

44. Consider the circuit shown where $C_1 = 6\mu F, C_2 = 3\mu F$ and V = 20 V. Capacitor C_1 is first charged by closing the switch S_1. Switch S_1 is then opened, and the charged capacitor is connected to the uncharged capacitor C_2 by closing S_2.



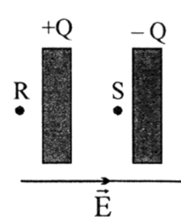
(A) Final total charge on C_1 and C_2 is $120\mu C$

(B) When only S_1 is closed heat produced is 12×10^{-4} J

(C) After S_2 is closed additional heat produced is 4×10^{-4} J

(D) Charge conversation does not hold when S_2 is closed

45. Two large thin conducting plates with small gap in between are placed in a uniform electric field 'E' (perpendicular to the plates). Area of each plate is A and charges + Q and – Q are given to these plates as shown in the figure. If points R, S and T as shown in the figure are three points in space, then the



(A) field at point R is E

(B) field at point S is E

(C) field at point T is $\left(E + \frac{Q}{\epsilon_0 A}\right)$

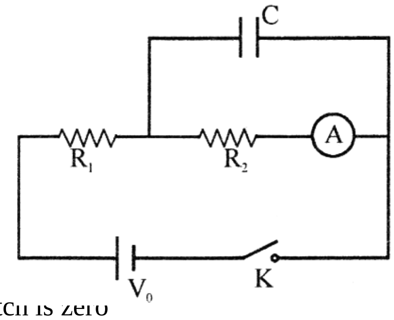
(D) field at point S is $\left(E + \frac{Q}{A\epsilon_0}\right)$

46. The current in a wire is doubled. Which of the following are correct statements?

(A) The current density is doubled **(B)** The conduction electron density is doubled

(C) The mean time between collisions is constant **(D)** The electron drift speed is doubled

47. In the connection shown in the figure initially the switch K is open and the capacitor is uncharged. Then the switch is closed and the capacitor is charged up to the steady state and the switch is opened again. [Given: $V_0 = 30$ V, $R_1 = 10\ k\Omega$, $R_2 = 5\ k\Omega$]



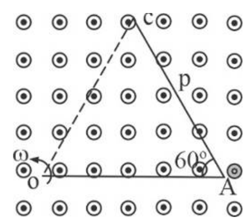
(A) Current in ammeter just after closing the switch is zero

(B) A long time after the switch was closed, the current in ammeter is 2mA:

(C) Just after reopening the switch, the current in ammeter is 2mA.

(D) Just after reopening the switch, the current in ammeter becomes zero

48. A metallic V shaped rod OAC is rotated with respect to one of its end in uniform magnetic field such that axis of rotation is parallel to the direction of magnetic field. Length of each arm of rod is L and angle between the arms is 60°. P is the mid-point of section AC. Magnitude of magnetic field is B. Then choose the correct relations.



(A) $V_A - V_0 = \frac{\omega BL^2}{2}$ **(B)** $V_A - V_C = \frac{\omega BL^2}{2}$

(C) $V_C - V_P = \frac{\omega BL^2}{8}$ **(D)** $V_A - V_P = \frac{\omega BL^2}{8}$

49. The electric potential at any point is given by $V = x\left(y^2 - 4x^2\right)$ volt. Then at point (1m,1m)

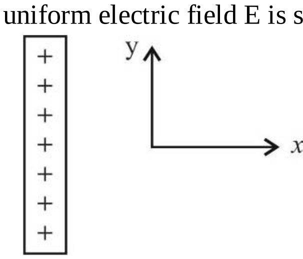
(A) x-component of electric field is 11 volt/m

(B) x-component of electric field is 4 volt/m

(C) y-component of electric field is –2 volt/m

(D) y-component of electric field is –4 volt/m

50. An insulating rod of uniform charge density λ and uniform linear mass density μ lies on a smooth table in horizontal xy-plane. A uniform electric field E is switched on :



(A) If electric field is along x-axis, the speed of the rod when it has travelled a distance D is $\sqrt{\frac{2\lambda ED}{\mu}}$

(B) If electric field E is at an angle $\theta(< 90°)$ with x-axis, the speed of the rod when it has travelled a distance D is $\sqrt{\frac{2\lambda EB\cos\theta}{\mu}}$

(C) Torque on the rod due to the field about centre of mass in case B is in the plane of paper

(D) Torque on the rod due to field about centre of mass in case B is zero

51. A conducting sphere of radius b has a spherical cavity with its centre (O_2) displaced by 'a' from centre of sphere (O_1). A point charge q is placed at the centre of cavity. Q charge is given to conducting sphere and charge q_0 is placed at a point P outside the sphere. P is at a distance c from centre of sphere such that O_1, O_2 and P are collinear.

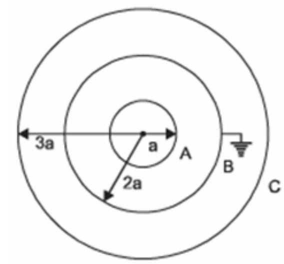
(A) Charge distribution on inner surface of cavity is uniform

(B) Potential of conductor is $\left(\frac{q_0}{4\pi\varepsilon_0 c}+\frac{Q+q}{4\pi\varepsilon_0 b}\right)$

(C) Charge distribution of outer surface of conducting sphere is non uniform

(D) Electric field inside the cavity is zero

52. Figure shows a system of three concentric metal shells A, B and C with radii a, $2a$ and $3a$ respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell A, then correct options are:



(A) $V_A = V_C$ and $V_B = 0$ **(B)** $q_1 = -\frac{q_2}{4}$

(C) $q_2 = -\frac{8}{11}Q$ **(D)** None of these

53. A long, hollow conducting cylinder is kept coaxially inside another long, hollow conducting cylinder of larger radius. Both the cylinders are initially electrically neutral.

(A) A potential difference appears between the two cylinders when a charge density is given to the inner cylinder

(B) Neglecting End effects, Electric Field lines are always perpendicular to curved surface of the two cylinders

(C) No potential difference appears between the two cylinders when a uniform line charge is kept along the axis of the cylinder

(D) Potential difference appears between the two cylinders when inner is earthed and outer is given some charge

54. A point charge q is placed at origin. Let $\vec{E}_A, \vec{E}_B$ and $\vec{E}_C$ be the electric field at three points A (1, 2, 3), B (1, 1, – 1) and C (2, 2, 2) due to charge q. Then :

(A) $\vec{E}_A \perp \vec{E}_B$ **(B)** $\vec{E}_A = \vec{E}_B$ **(C)** $\left|\vec{E}_B\right| = 4\left|\vec{E}_C\right|$ **(D)** $\vec{E}_B = 16\left|\vec{E}_C\right|$

55. An inductor 4H and a resistance 5Ω are connected in series with an AC source. At a particular instant, voltage across inductor is 3 volt and across resistor is 4 volt. For that particular instant, choose correct options :

(A) Voltage across source is 5 volt **(B)** Voltage across source may be 7 volt

(C) Voltage across source may be 1 volt **(D)** Current in circuit is 0.8 amp

56. Two alike discharge lamps are operated from the mains, 230 V and 50 Hz, and they are both connected to inductors in series. The inductors are alike. A condenser is connected in series to one of the discharge tubes. With this arrangement it is ensured that when the current through one of the lamps is maximum, then the current through the other one is minimum and vice-versa. The average value of the power is 8W for each discharge lamp. Then which of the following is(are) correct ? (Assume discharge lamps as resistors)

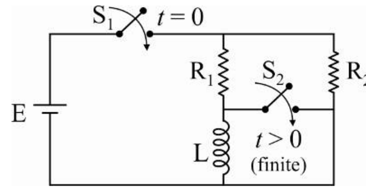
(A) The value of resistance of discharge tube is 3.3 $k\Omega$

(B) The capacitance of condenser is 6.6 μF

(C) The total impedence of the discharge tube and the inductor together is $4.67\,k\Omega$

(D) The capacitive reactance is double of inductive reactance.

57. In the circuit shown below, the current through inductor is zero when battery is not connected ($t < 0$). At $t = 0$, switch S_1 is closed. If at a later time $t = t_1$ (finite), switch S_2 is also closed, then which of the following is true just after time t_1 ?



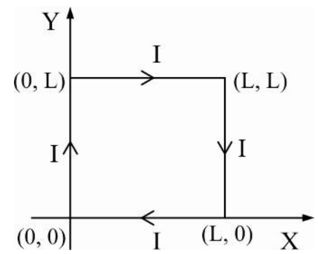
(A) Current through resistor R_1 increases

(B) Current through resistor R_2 increases

(C) Current through resistor L increases

(D) Current through the battery increases

58. Figure shows a square loop in x-y plane carrying current I present in the magnetic field which is given by $\vec{B} = \frac{B_0 z}{L}\hat{j} + \frac{B_0 y}{L}\hat{k}$ where B_0 is positive constant, which of the following statement(s) is(are) correct ?



(A) Force on side (0, 0) to (0, L) is $\left(\frac{B_0 IL}{2}\right)\hat{i}$

(B) Force on side (0, 0) to (0, L) is $-B_0 IL\,\hat{j}$

(C) Net magnetic force on loop is zero

(D) Force on side (L, 0) to (0, 0) is zero

59. A long cylindrical conductor whose axis is coincident with the z-axis has an internal magnetic field given by $B = \frac{\mu_0 J_0 r}{6a}(3a - 2r)$ where a is the conductor's radius.

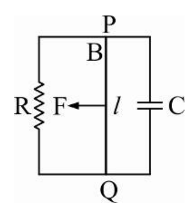
(A) The total current flowing in the conductor is $\frac{J_0 \pi a^2}{3}$

(B) The magnetic field outside the conductor is $\frac{\mu_0 J_0 \pi a^2}{6r}$

(C) The current density is maximum at the axis of the conductor

(D) The current density is minimum at the axis of the conductor

60. A conducting rod PQ of length l is pulled with a constant force F along two smooth parallel rails separated by a distance l as shown in the adjacent figure. Magnetic field B is perpendicular to the plane. Then choose the correct statement(s).



(A) Terminal velocity of the rod, $v_t - \frac{2FR}{B^2 l^2}$

(B) Terminal velocity of the rod, $v_t = \frac{RF}{B^2 l^2}$

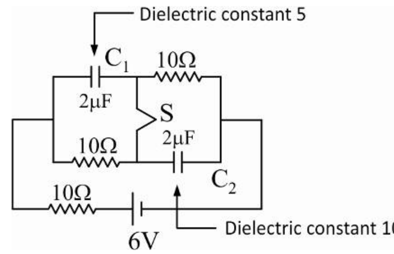
(C) Maximum charge on the capacitor, $q_{max} = \frac{FCR}{Bl}$

(D) Maximum charge on the capacitor, $q_{max} = \frac{2FCR}{Bl}$

SECTION-4	COLUMN MATCHING TYPE

Answer Q.61, Q.62 and Q.63 by appropriately matching the information given in the three columns of the following table.

Two capacitors C_1 and C_2 each have capacitance of 2 μF. Consider each previous question as a permanent step for the next question. Values of work done by battery, heat developed and $|Q_2 - Q_1|$ are given for that question till system again achieves steady state. Initially the system is in steady state and switch S is in closed state.



Column I		Column II		Column III	
Work done by the battery in that step (in μJ)		Heat developed in the circuit (in μJ) in that step		$\|Q_2 - Q_1\|$ where Q_2 and Q_1 are final charge on capacitor C_2 and C_1 (in μF) in that step	
I.	360	**(i)**	180	**(P)**	36
II.	216	**(ii)**	140	**(Q)**	60
III.	720	**(iii)**	80	**(R)**	20
IV.	96	**(iv)**	240	**(S)**	16

61. A dielectric of dielectric constant 5 is inserted in the capacitor C_1 :

(A) (IV) (iii) (S) **(B)** (II) (i) (S) **(C)** (III) (iv) (Q) **(D)** (I) (ii) (R)

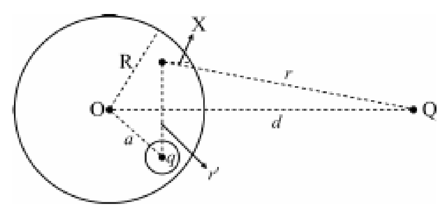
62. A dielectric of dielectric constant 10 is inserted in the capacitor C_2 :

(A) (III) (iv) (Q) **(B)** (II) (i) (R) **(C)** (IV) (iii) (P) **(D)** (II) (ii) (Q)

63. Now switch S is opened :

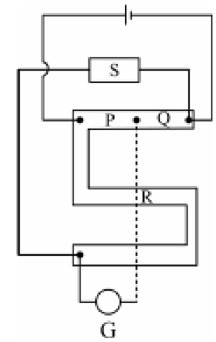
(A) (I) (ii) (Q) **(B)** (II) (i) (S) **(C)** (III) (IV) (Q) **(D)** (IV) (iii) (P)

64. A charge q is placed inside a spherical cavity at its centre, made in an uncharged conducting sphere of radius R as shown. A point charge Q is placed at a separation d from the centre of solid sphere as shown in the figure.



	Column I		Column II
(A)	Electrostatic potential at point X inside the conductor	**(p)**	$\frac{kQ}{d}+\frac{kq}{R}-\frac{kq}{r'}-\frac{kQ}{r}$
(B)	Electrostatic potential at point O	**(q)**	$\frac{kQ}{r^2}$
(C)	The magnitude of the electrostatic field at point X inside the conductor due to the charges induced at the outermost surface only	**(r)**	$\frac{kQ}{r}$
(D)	Electrostatic potential due to all the induced charges at point X inside the conductor	**(s)**	$\frac{kQ}{d}+\frac{kq}{R}$
		(t)	$\frac{kQ}{2d}+\frac{kq}{2R}$

65. Connections made in a post office box are shown in figure. $R\nearrow = 10\Omega$ denotes that when R = 10Ω the pointer in the galvanometer is as $\nearrow$.

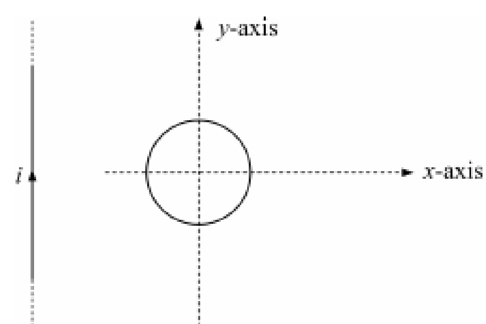


Column I		Column II	
(A)	P = 100Ω, Q = 10Ω, $R\nearrow = 460\Omega$, $R\nwarrow = 500\Omega$	**(p)**	$46\Omega < S < 47\Omega$
(B)	P = 100Ω, Q = 1Ω, $R\nearrow = 460\Omega$, $R\nwarrow = 470\Omega$	**(q)**	$0.46\Omega < S < 0.47\Omega$
(C)	P = 100Ω, Q = 10Ω, $R\nearrow = 460\Omega$, $R\nwarrow = 470\Omega$	**(r)**	$46\Omega < S < 50\Omega$
(D)	P = 1000Ω, Q = 1Ω, $R\nearrow = 460\Omega$, $R\nwarrow = 470\Omega$	**(s)**	$4.6\Omega < S < 4.7\Omega$

66. MATCH THE FOLLOWING :

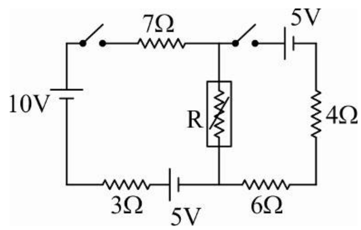
Column I		Column II	
(A)	Plates of an isolated, charged, parallel plate, air cored capacitor are slowly pulled apart	**(p)**	Electric energy stored inside capacitor increases in the process
(B)	A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor	**(q)**	Force between the two plates of the capacitor remains unchanged
(C)	Plates of a parallel plate air cored capacitor connected across a battery are slowly pulled apart	**(r)**	Electric field in the region between plates remains unchanged
(D)	A dielectric slab is slowly inserted inside an air cored parallel plate capacitor connected across a battery to completely fill the space between plates	**(s)**	Total electric energy stored inside capacitor decreases in the process
		(t)	Electric field in the region between plates decreases

67. A conducting circular rigid loop near a long straight current carrying wire as shown. Match the following table.



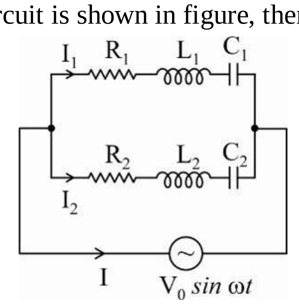
Column I		Column II	
(A)	If current is increased	**(p)**	Induced current in the loop is clockwise
(B)	If current is decreased	**(q)**	Induced current in the loop is anticlockwise
(C)	If loop is moved away from the wire maintaining constant current in the straight wire	**(r)**	Wire will attract the loop and there will be no torque about y-axis
		(s)	Wire will repel the loop and there will be no torque about y-axis

68. In the circuit shown the resistance R is kept in a chamber whose temperature is $20°C$ which remains constant. The initial temperature and resistance of R is $50°C$ and 15Ω respectively. The rate of change of resistance R with temperature is $\frac{1}{2}\frac{\Omega}{°C}$ and rate of decrease of temperature of R is $\frac{(ln 3)}{100}$ times the temperature difference from surrounding. Assume the resistance R loses heat only in accordance with Newton's law of cooling. If both keys are closed at $t = 0$.



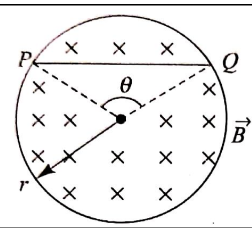
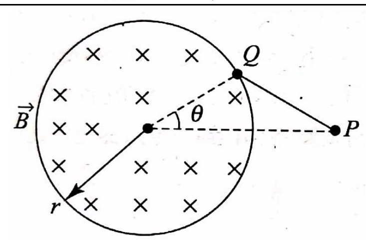
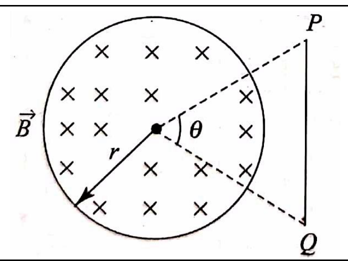
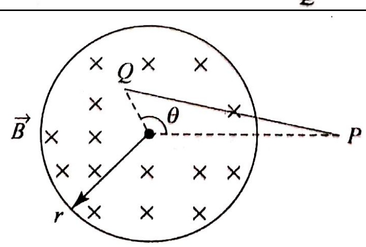
	Column I		Column II
(A)	The value of R (in Ω) for which power dissipation in it is maximum is	**(p)**	6
(B)	Temperature of R when power dissipation in it is maximum is $T°C$ then $\frac{T}{5}$ is	**(q)**	10
(C)	Time after which power dissipation in it is maximum is t sec then $\frac{t}{10}$ is	**(r)**	5
(D)	Current (in Amp.) through 10 V battery when power dissipation in R is maximum is	**(s)**	1

69. An alternating LCR circuit is shown in figure, then match the column :



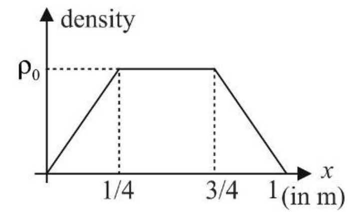
	Column I		Column II
(A)	If $\omega L_1 - \frac{1}{\omega C_1} = R_1$ and $\omega L_2 - \frac{1}{\omega C_2} = R_2$	**(p)**	I_1 and I_2 are in same phase
(B)	If $\omega L_1 - \frac{1}{\omega C_1} = R_1$ and $\frac{1}{\omega C_2} - \omega L_2 = R_2$	**(q)**	$I = I_1 + I_2$ (I, I_1, I_2 rms value of currents)
(C)	If capacitor is absent in 1 and inductor is absent in 2 and $\omega L_1 = R_1$, $\frac{1}{\omega C_2} = R_2$	**(r)**	Phase difference between I_1 and I_2 is $\frac{\pi}{2}$
(D)	If capacitor are not present in both and $\omega L_2 = R_2$, $\omega L_1 = R_1$	**(s)**	$I = I_1 + I_2$ (I, I_1, I_2 rms value of currents)

70. Column I shows the cylindrical region of radius r where a inward magnetic field $\vec{B}$ exists, where $\vec{B}$ is increasing at the rate of dB/dt. A rod PQ is placed in different position as shown. Match the column I with correct statement in column II regarding the induced emf in rod.

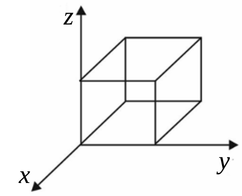
	Column I		Column II
(A)		**(p)**	Induced emf in rod PQ is $\frac{1}{2}r^2\theta\frac{dB}{dt}$
(B)		**(q)**	Induced emf in rod PQ is less than $\frac{1}{2}r^2\theta\frac{dB}{dt}$
(C)		**(r)**	End P is positive with respect to end Q.
(D)		**(s)**	End Q is positive with respect to end P.

SECTION-5	NUMERICAL TYPE

71. The volume charge density as a function of distance X from one face inside a unit cube is varying as shown in the figure. Then the total flux (in S.I. units) through the cube if $\rho_0 = 8.85 \times 10^{-12}\ C/m^3$ is ________.



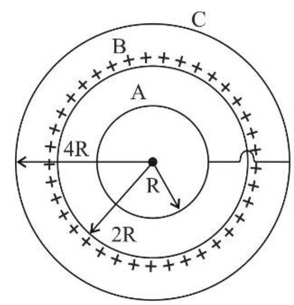
72. Figure below shows a closed Gaussian surface in the shape of a cube of edge length 3.0 m. There exists an electric field given by $\vec{E} = \left[(2.0x + 4.0)i + 8.0\,j + 3.0\,k\right] N/C$, where x is in metres, in the region in which it lies. The net charge in coulombs enclosed by the cube is $n \in_0$, where n is __________.



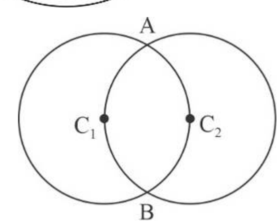
73. A charged particle of charge 'Q = 5C' is held fixed and another charged particle of mass '$m = 5\pi\ kg$' and charge '$q = 4\varepsilon_0$ C' (of the same sign) is released from a distance '$r = 1\ m$'. The impulse (in Ns) of the force exerted by the external agent on the fixed charge by the time distance between 'Q' and 'q' becomes 2 m is___________.

74. A thin spherical steel shell has radius $R = \frac{1}{2}m$ and thickness of 0.09mm. It is given a charge Q. What can be the maximum charge given to the shell so that it may not break due to electrostatic repulsion ? Breaking strength for steel $= 8 \times 10^8$ N/m^2. If answer is $N\sqrt{\pi} \times 10^{-3} C$, find N.

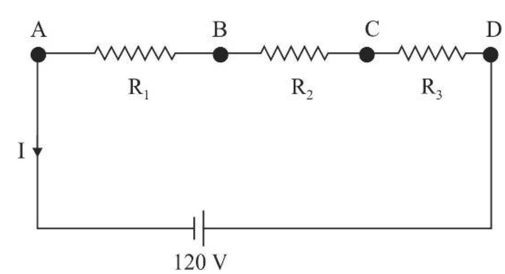
75. Three concentric spherical conductors A, B and C of radii R, 2R and 4R respectively. A and C is shorted and B is uniformly charged (charge + Q). Charge on conductor A is –Q/X where X is ___________.



76. Two circular uniform rings of identical radii and resistance of $36\,\Omega$ each are placed in such a way that they cross each other's center C_1 and C_2 as shown in figure. Conducting joints are made at intersection points A and B of the rings. An ideal cell of emf 20 V is connected across A and B. The power delivered by the cell is $x \times 10^2$ watt. Find the value of x.

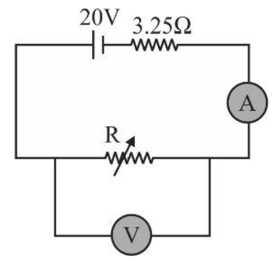


77. In the circuit shown potential differences $V_C - V_A = 60$ Volt and $V_D - V_B = 90$ Volt. The internal resistance of battery is 20Ω and I = 0.5 A. If the resistance R_2 (in ohm) is 10X. Then find the value of X.



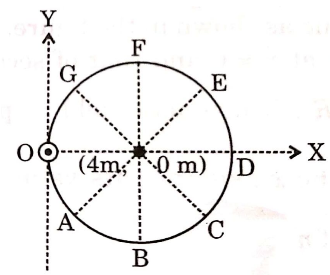
78. A galvanometer of resistance R_G is to be converted into an ammeter, with the help of a shunt resistance R. If the ratio of heat dissipated through galvanometer and shunt is 3 : 4 when connected in circuit, then R is X times R_G where X is __________.

79. A Galvanometer G of coil resistance 15Ω having 30 equal divisions and full scale deflection corresponding to 60 mA current is converted into ammeter by using 5Ω shunt. This ammeter is connected in circuit shown in figure. Voltmeter in circuit is ideal voltmeter. The value of R for which deflection in ammeter is corresponding to 10 division, is __________ (in Ω).

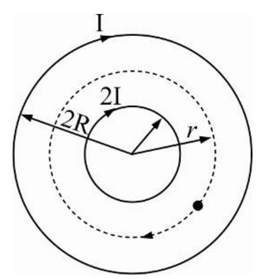


80. In a potentiometer experiment the balancing with a cell is at length 240 cm. On shunting the cell with a resistance of 2Ω, the balancing length becomes 120 cm. The internal resistance of the cell is __________ (in Ω).

81. An infinite uniform current carrying wire is kept along z-axis, carrying current I_0 in the direction of the positive z-axis. OABCDEFG represents a circle (where all the points are equally spaced), whose centre at point (4m, 0m) and radius 4m as shown in the figure. If $\int \vec{B} \cdot \vec{d}\,l = \frac{\mu_0 I_0}{k}$ in S.I. unit, then find the value of k.

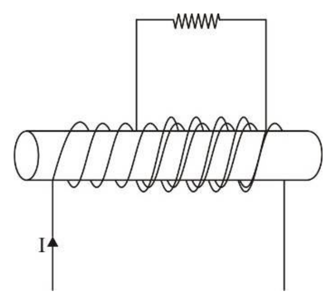


82. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have the same number of turns per unit length and initially both carry no current. At the same instant current starts increasing linearly with time in both solenoids. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing currents a charged particle, initially at rest between the solenoids, starts moving along a circular trajectory of radius r (see figure). The value of $\frac{r\sqrt{2}}{R}$ is __________.



83. Between the plates of a parallel-plate capacitor having separation d, we put a metallic plate of thickness 0.6 d. When that plate is absent the capacitor has a capacity $c = 20nF$. The capacitor is connected to a dc voltage source of emf of 100 volt. The metallic plate is slowly extracted from the gap. Find the mechanical work performed in the process of plate extraction (in μJ). If your answer is N find value of N/30.

84. A coil with 1500 turns, a radius of 5.0 cm and a resistance of 12Ω surrounds a solenoids with 240 turns/cm and a radius of 4 cm as shown in figure. The current in the solenoid changes at a constant rate from 0 to 20A in 0.10 s. Calculate the magnitude of the induced current (in Amp, upto 1 decimal place) in the 1500 turn coil ($\pi^2 = 10$, neglect self inductance of the coil).

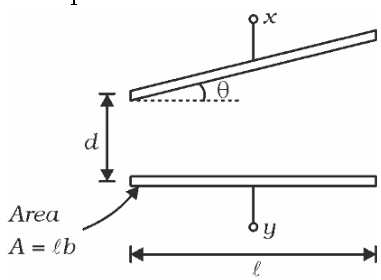


85. Two boys are holding a wire, standing $4m$ apart. The wire stays in the shape of a circular arc with a sag of $1m$. The students rotate the wire about the horizontal line connecting their hands, as if they were playing jump rope. The boys rotate the wire at a speed of 4 revolutions per second. The earth's magnetic field at the location is $4\times10^{-5}\,T$. Calculate the RMS value of emf developed between the ends of the wire (in mV, to nearest integer). Assume the shape of the wire is maintained as it is rotated.

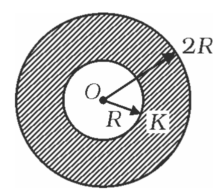
SECTION-6	SUBJECTIVE TYPE

86. A particle of mass m and charge q starts from origin at $\vec{v}=v_0\hat{i}$ in a magnetic field $\vec{B}=B_0x\hat{k}$. Find the maximum x co-ordinate of particle in the subsequent motion of particle.

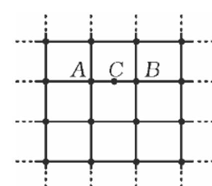
87. Figure shows a capacitor with one plate slightly tilted by a very small angle θ to the other plate. Find the capacitance of this capacitor.



88. Figure shows a metal sphere of radius R which is enveloped by a dielectric layer of radius $2R$. Find the capacitance of this sphere.



89. In an infinite grid of square cells made of conducting wires, resistance of an edge of a square cell is R. In the figure, point C is midpoint of the edge AB. Find resistance of the entire grid as measured between the terminal pairs A-B and A-C.



90. In a uniform magnetic field B the nucleus of mass a neutral atom of number $100m$ emits an α-particle of mass $4m$. Find the time after emission when α-particle will collide with the daughter atoms formed after emission. (Neglect electrostatic interactions)

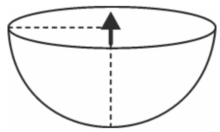
91. A resistance (R), Inductance (L) and capacitance (C) are connected in series to an ac source of voltage V having variable frequency. Calculate the energy delivered by the source to the circuit during one period if the operating frequency is twice the resonance frequency.

92. Each of two long parallel wires carries a constant current I along the same direction. The wires are separated by a distance $2l$. Calculate maximum magnitude of magnetic induction in the symmetry plane of this system located between the wires. Calculate also, the maximum force experienced by unit length of a third wire carrying the same current along the same direction if the third wire is parallel to and in the symmetry plane of other two wires.

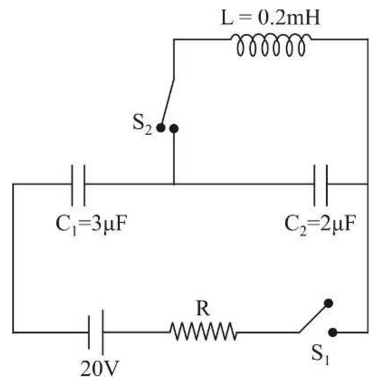
93. A parallel capacitor is filled with a faulty dielectric of dielectric constant k and it also has some resistivity ρ. If a charge Q is given to the capacitor, it discharges through the medium between plates. Find time constant of discharging of this capacitor and the leakage current in capacitor.

94. Two long parallel wires of negligible resistance are connected at one end to a resistance R and at the other end to a constant voltage source of voltage V. The distance between the axes of the wire is η times greater than cross-sectional radius of each wire. At what value of resistance R, does the resultant force of interaction between the wires will become zero?

95. A small freely oriented electric dipole of dipole moment $\vec{P}$ is placed at the centre of a charged hemispherical cup of surface charge density σ coul/m^2 with dipole moment vector along axis of cup as shown in figure. Find the period of small oscillations of dipole about diametrical axis of base of cup. Moment of inertia of dipole about the given axis is I.



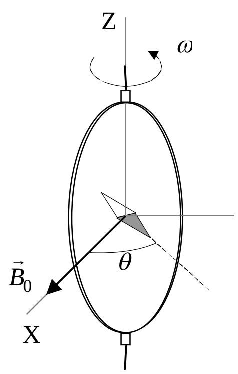
96. The circuit shown in the figure is in the steady state with switch S_1 closed. At t = 0, S_1 is opened and switch S_2 is closed.



(i) Derive an expression for the charge on the capacitor C_2 as function of time.

(ii) Determine the first instant t, when the energy in the inductor becomes one-third of that in the capacitor.

97. A closed circular coil of N turns, radius a and total resistance R is rotated with uniform angular velocity ω about a vertical diameter in a horizontal magnetic field $\vec{B}_0 = B_0\,\vec{i}$.



(i) Compute the electromotive force ε induced in the coil, and also the mean power $\langle P \rangle$ required for maintaining the coil in motion. Neglect the coil self inductance.

(ii) A small magnetic needle is placed at the center of the coil, as shown in figure. It is free to turn slowly around the Z axis in a horizontal plane, but it cannot follow the rapid rotation of the coil. Once the stationary regime is reached, the needle will set at a direction making a small angle θ with $\vec{B}_0$. Compute the resistance R of the coil in terms of this angle and the other parameters of the system.

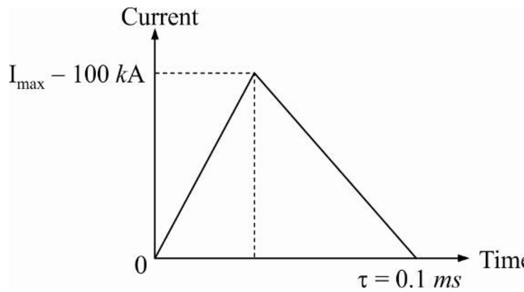
98. Output of a faulty inverter is expressed for one cycle as $emf = E(\theta) = \begin{cases} \dfrac{E_m}{\alpha}\theta, & 0 < \theta \le \alpha \\ E_m, & \alpha \le \theta \le \pi - \alpha \\ \dfrac{-E_m}{\alpha}\theta, & \pi - \alpha \le \theta \le \pi \end{cases}$

(i) Find the ratio of average value of emf when $\alpha = \dfrac{\pi}{6}$ and when $\alpha = \dfrac{\pi}{2}$.

(ii) For $\alpha = 0,$ find the rms value of emf.

(iii) Find the ratio of rms values for $\alpha = \dfrac{\pi}{6}$ and for $\alpha = \dfrac{\pi}{2}$.

99. An oversimplified model of lightning is presented. Lightning is caused by the build-up of electrostatic charge in clouds. As a consequence, the bottom of the cloud usually gets positively charged and the top gets negatively charged, and the ground below the cloud gets negatively charged.



When the corresponding electric field exceeds the breakdown strength value of air, a disruptive discharge occurs; this is lighning, Idealized current pulse flowing between the cloud and the ground during lightning is as shown in the figure.

Answer the following questions with the aid of this simplified curve for the current as a function of time and these data :
Distance between the bottom of the cloud and the ground : $h = 1$ km;

Breakdown electric field of humid air : $E_0 = 300 \;\; kV\,\mathrm{m}^{-1}$;

Total number of lightning striking Earth per year : 32×10^6;

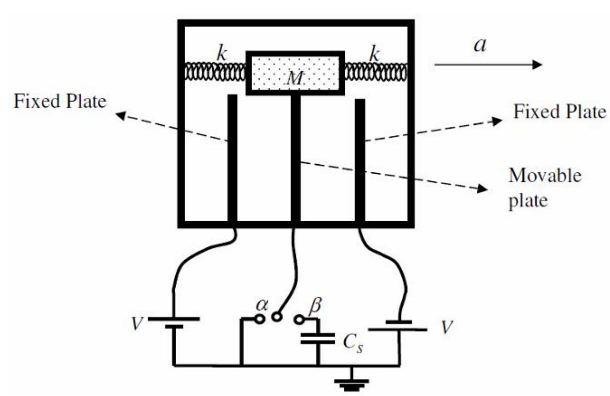
Total human population : 6.5×10^9 people.

(i) What is the total charge Q released by lightning ?

(ii) What is the average current I flowing between the bottom of the cloud and the ground during lightning?

(iii) Imaging that the energy of all storms of one year is collected and equally shared among all people. For how long could you continuously light up a 100 W light bulb for your share ?

100. Figure shows a mass M which is attached to a conducting plate with negligible mass and also to two springs having identical spring constants k. The conducting plate can move back and forth in the space between two fixed conducting plates. All these plates are similar and have the same area A. Thus these three plates constitute two capacitors.



As shown in figure, the fixed plates are connected to the given potentials V and $-V$, and the middle plate is connected through a two-state switch to the ground. The wire connected to the movable plate does not disturb the motion of the plate and the three plates will always remain parallel. When the whole complex is not being accelerated, the distance from each fixed plate to the movable plate is d which is much smaller than the dimensions of the plates. The thickness of the movable plate can be ignored.

The switch can be in either one of the two states α and β. Assume that the capacitor complex is being accelerated along with a automobile, and the acceleration is constant. Assume that during this constant acceleration the spring does not oscillate and all components of this complex capacitor are in their equilibrium positions, i.e. they do not move with respect to each other, and hence with respect to the automobile.

Due to the acceleration, the movable plate will be displaced a certain amount x from the middle of the two fixed plates. (Assume $d >> x$)

(i) Consider the case where the switch is in state α i.e. the movable plate is connected to the ground through a wire, then

- **(a)** Find the charge on each capacitor as a function of x.
- **(b)** Find the net electric force on the movable plate, F_E, as a function of x.
- **(c)** Express the constant acceleration a as a function of x.

(ii) Now assume that the switch is in state β i.e. the movable plate is connected to the ground through a capacitor, the capacitance of which is C_s (there is no initial charge on the capacitors). If the movable plate is displaced by an amount x from its central position, find V_s the electrical potential difference across the capacitor C_s as a function of x.